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VEHICLE EQUIPMENT MOUNT

Abstract

A vehicle equipment mount for a vehicle having a tow hook includes a clamp configured to mount on the tow hook, the clamp including a mounting surface for affixing vehicle equipment.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION [0001] This application claims priority to U.S. Provisional Application, Ser. No. 63/563,743, filed on 11 Mar. 2024. This U.S. Provisional Application is hereby incorporated by reference herein in its entirety and is made a part hereof, including but not limited to those portions which specifically appear hereinafter.

BACKGROUND OF THE INVENTION

Field of the Invention

[0002] This invention relates generally to a mounting system for vehicle equipment, specifically a truck brush guard.

Description of Prior Art

[0003] There is a substantial interest in the automotive aftermarket for modifications and similar improvements to vehicle appearance, utility and/or performance. More specifically, a need exists for automotive equipment that can be added to factory vehicles such as trucks and SUVs.

[0004] Often such aftermarket equipment requires substantial mechanical modifications, including welding and complex subframe assemblies, to facilitate additions such as racks, brush guards, tow packages and similar. Such substantial modifications may deter the casual user resulting in a potential loss of market.

[0005] A need therefore exists in the automotive aftermarket, specifically in trucks and SUVs, for a simple, safe and reliable modification platform to facilitate the addition of aftermarket equipment. Although the focus of the subject invention is on aftermarket equipment, the subject invention is equally applicable to the original equipment manufacturer (OEM).

SUMMARY OF THE INVENTION

[0006] The invention generally relates to a mounting system that utilizes OEM tow hooks as a platform for mounting aftermarket equipment, such as a brush guard.

[0007] Other objects and advantages will be apparent to those skilled in the art from the following detailed description taken in conjunction with the appended claims and drawings.

Description

BRIEF DESCRIPTION OF THE DRA WINGS

[0008] FIG. **1** shows a schematic of a vehicle according to one preferred embodiment;

[0009] FIG. **2** shows an exploded top view of a clamp according to one preferred embodiment;

[0010] FIG. **3** shows a front perspective view of a tow hook and fastener plate according to one preferred embodiment;

[0011] FIG. **4** shows a front perspective view of a tow hook, cover, and fastener plate according to one preferred embodiment;

[0012] FIG. **5** shows a front perspective view of a tow hook and assembled clamp according to one preferred embodiment;

[0013] FIG. **6** shows a front perspective view of a tow hook and clamp with vehicle equipment according to one preferred embodiment;

[0014] FIG. **7** shows a top view of a support strap and insulating nut according to one preferred embodiment;

[0015] FIG. **8** shows a front perspective view of a support strap connected to a vehicle and the vehicle equipment according to one preferred embodiment;

[0016] FIG. **9** shows a side perspective view of a brush guard according to one preferred embodiment;

[0017] FIG. **10** shows a front view of the brush guard shown in FIG. **9**;

[0018] FIG. **11** shows a side perspective exploded view of a clamp according to one embodiment; and

[0019] FIG. **12** shows a bottom view of a cover shown in the clamp in FIG. **11**.

DETAILED DESCRIPTION OF THE INVENTION

[0020] The present invention provides a system for mounting vehicle equipment **100** to a vehicle **5** such as shown schematically in FIG. **1**. Vehicle equipment **100** may include such things as a brush guard **110**, light bar, storage rack, winch, hitch mount, flag mount, bull bars, bike rack, spare tire holder, auxiliary lighting, jack mount, a luggage rack and/or other mountable vehicle components, accessories, and mounts. As shown in the drawings, and for the purposes of the specification, the following description is directed to an embodiment including a brush guard **110** although any of the above options is represented schematically in FIG. **1**. It is noted that the described system may be configured for attachment to the front and/or rear of a vehicle **5**. The subject brush guard **110** embodiment is intended for use in the front of the vehicle **5**.

[0021] Many vehicles, particularly heavy duty and work use vehicles such as pickup trucks, include integrated tow hooks **10** that are generally used to attach chains or tow straps for use to tow the vehicle or another out of snow, mud or terrain. Such tow hooks **10** are also used to pull or move heavy objects, again with a chain, rope or strap. Lastly, such tow hooks **10** may be used by towing vehicles to winch the vehicle onto a flatbed or into position for a tow. These tow hooks **10** may be oriented vertically or horizontally and may be attached to the front and/or rear of the vehicle. One commonality of all applications is that the tow hook **10** includes some sort of central aperture **15**. In addition, tow hooks **10** are generally anchored to a frame of the vehicle to provide sufficient strength and rigidity for the various requirements.

[0022] As shown in FIGS. **1-12**, the subject system may include one and preferably two clamps **20** each for mounting to a corresponding tow hook **10** in the front or rear of the vehicle. As shown in FIG. **2**, the clamps **20** preferably include a two-piece clamshell design that sandwiches around a corresponding tow hook **10** and further includes a mounting surface and/or a mounting aperture for affixing vehicle equipment **100**.

[0023] As shown in the figures, the vehicle equipment mount includes a pair of clamps **20** and a pair of tow hooks **10**, so that each clamp **20** of the pair of clamps **20** mounts to a separate tow hook **10**. As shown in FIG. **2**, each clamp **20** preferably includes a cover **30** and a fastener plate **40** wherein the cover **30** and the fastener plate **40** form a clamshell around the tow hook **10**. In addition, each clamp **20** may further include a contact plate **55**.

[0024] FIGS. **3** and **4** show one preferred embodiment of the clamp **20** wherein the cover **30** overlays the tow hook **10** and the fastener plate **40** fits through or within the aperture **15** of the tow hook **10**. One or more fastening bolts may be used to connect the fastener plate **40** to the cover **30** around and/or through the tow hook **10**. As shown in FIG. **5**, a contact plate **55** may be attached to the cover **30** and may fit flush within a shallow pocket on the cover **30**. The contact plate **55** preferably provides a smooth finished surface for affixing the vehicle equipment **100**.

[0025] In addition, and as best shown in FIGS. **2**, **11** and **12**, the fastener plate **40** may be contoured to fit within a cavity **35** in the cover **30**. Preferably this contour is asymmetrical across the fastener plate **40** such that the fastener plate **40** is engageable and connectable with the cover **30** in only a single alignment. This arrangement prevents user error in attaching the clamps **20** to the tow hooks **10** by ensuring correct placement and alignment of each clamp **20**.

[0026] FIG. **6** shows attachment of the vehicle equipment **100**, in this case a brush guard **110**, with a connecting bolt **75**. To accommodate this attachment, the cover **30** and/or the fastener plate **40** and/or the contact plate **55** include a through hole **70** configured to mount the vehicle equipment **100** such as with one or more bolts **75** or other connectors.

[0027] In the embodiment shown in the drawings wherein a brush guard **110** is attached to the vehicle, and as specifically shown in FIGS. **7** and **8**, at least one support strap **60** is used to provide a second axis of connection between the brush guard **110** and the vehicle. The vehicle equipment **100**, such as brush guard **110**, is preferably attached at each clamp **20** and further attached to a second mounting point on the vehicle. As such, the support strap **60** may extend between an upper

surface of the brush guard **110** and a second mounting point on the vehicle, specifically within the engine bay as shown in FIG. **8**. In addition, the support strap **60** may be attached at the second mounting point with a insulating nut **65** as shown in FIG. **7**. The insulating nut **65** preferably includes a rubberized grommet or insert to isolate the attachment point and reduce possible vibration.
[0028] According to one embodiment of the subject invention (not shown), the vehicle equipment mount may include a single structural clamp **20** that extends between and affixes to two tow hooks **10**. Such an arrangement may be helpful for accessories or equipment that requires an even greater structural support.
[0029] The invention illustratively disclosed herein suitably may be practiced in the absence of any element, part, step, component, or ingredient which is not specifically disclosed herein.
[0030] While in the foregoing detailed description this invention has been described in relation to certain preferred embodiments thereof, and many details have been set forth for purposes of illustration, it will be apparent to those skilled in the art that the invention is susceptible to additional embodiments and that certain of the details described herein can be varied considerably without departing from the basic principles of the invention.

Claims

1. A vehicle equipment mount for a vehicle having a tow hook, the vehicle equipment mount comprising: a clamp configured to mount on the tow hook, the clamp including a mounting surface for affixing vehicle equipment.
2. The vehicle equipment mount of claim 1 further comprising at least one support strap extending between the vehicle equipment and a second mounting point on the vehicle.
3. The vehicle equipment mount of claim 1 comprising a pair of clamps and a pair of tow hooks, each clamp of the pair of clamps mounting to a separate tow hook.
4. The vehicle equipment mount of claim 1 wherein the clamp comprises a cover and a fastener plate wherein the cover and the fastener plate form a clamshell around the tow hook.
5. The vehicle equipment mount of claim 4 wherein the cover overlays the tow hook and the fastener plate fits within an aperture of the tow hook.
6. The vehicle equipment mount of claim 5 further comprising at least one bolt connecting the fastener plate to the first portion around and through the tow hook.
7. The vehicle equipment mount of claim 4 wherein the cover and the fastener plate include a through hole configured to mount the vehicle equipment.
8. The vehicle equipment mount of claim 4 wherein the fastener plate is contoured to fit within a cavity in the cover.
9. The vehicle equipment mount of claim 8 wherein the contour is asymmetrical across the fastener plate and is configured to engage with the cover in a single alignment.
10. The vehicle equipment mount of claim 1 further comprising a brush guard mounted to the clamp.
11. The vehicle equipment mount of claim 10 further comprising a strap connecting a top portion of the brush guard to the vehicle.
12. The vehicle equipment mount of claim 1 wherein the vehicle equipment comprises at least one of a winch, a hitch mount, a flag mount, bull bars, a bike rack, a spare tire holder, auxiliary lighting, a jack mount, and a luggage rack.
13. The vehicle equipment mount of claim 1 wherein the clamp affixes to two tow hooks.
14. A vehicle equipment mount for a vehicle having a tow hook, the vehicle equipment mount comprising: a pair of clamps configured to mount on a pair of tow hooks, the clamps each including a mounting surface for affixing vehicle equipment, a cover, and a fastener plate wherein the cover and the fastener plate form a clamshell around the tow hook; and at least one support

strap extending between the vehicle equipment and a second mounting point on the vehicle.
15. The vehicle equipment mount of claim 14 wherein the cover overlays the tow hook and the fastener plate fits within an aperture of the tow hook.
16. The vehicle equipment mount of claim 15 further comprising at least one bolt connecting the fastener plate to the first portion around and through the tow hook.